AUTOMATION CASE STUDY

Leasing and tenant screening

Five n8n workflows connect renter inquiries, tours, application checks, manager decisions and a sandbox lease.

The problem

Leasing work moves between chat, calendars, uploaded documents and a manager. Re-entering facts and chasing the next step makes it easy to lose context.

What I built

A connected coordination system: property-grounded replies, manager-approved tour booking, cited paystub extraction, deterministic criteria checks and a recorded approval path to signing.

Evidence from a fresh connected run

Maya Restrepo moved from inquiry R-9412 to application A-9731, a signed nonbinding sandbox lease and a received welcome email. A second synthetic application demonstrated the stop-for-human branch.

All applicants and property records in this sample are synthetic. Plaid uses canned Sandbox payroll data. Bureau, identity and entry-code outputs are labelled MOCK.

SYSTEM DESIGN

One record, connected workflows

n8n orchestrates; Postgres holds the lead, application, approvals, messages and state changes.

WF1 / Conversation

Listing inquiry, Telegram and email. Grounded facts and fixed prequal gates.

WF5 / Tours

Cal.com slots, manager approval, booking and rescheduling.

WF2 / Application

Form + paystub, OpenAI extraction, Sandbox payroll and fixed criteria.

WF3 / Manager router

Telegram buttons and commands route approved actions and record the actor.

Postgres / System of record

Reference tokens connect the journey. Airtable mirrors phase boundaries.

WF4 / Lease + welcome

Fixed document, DocuSign Sandbox callback and separate welcome approval.

Swappable MOCK adapters

Synthetic listing feed, bureau / eviction results, identity checks and entry codes. Dashed box denotes mock components; external calls to Cal.com, Plaid Sandbox and DocuSign Sandbox are genuine.

FRESH DEMONSTRATION

Follow one renter through the build

Captured on October 10, 2026. Reference tokens connect the screenshots and execution records.

1 / Inquiry and conversation

Maya asks about parking and pets. The assistant uses property facts and collects self-reported prequalification details.

2 / Tour approval and application

The manager approves October 12 at 11:00 AM property time. Cal.com creates the booking. A manager then records synthetic demo attendance and authorizes the application follow-up.

3 / Verified facts and a human decision

The uploaded synthetic paystub and Sandbox payroll both normalize to $4,996.33/month. Fixed criteria produce evidence; the manager separately approves the application.

4 / Signing and welcome

DocuSign completes the nonbinding sandbox document. The callback updates the lease and unit. A second manager approval releases the welcome email, which was received in the demo mailbox.

WORKFLOW 01

Renter intake and conversation

Actual n8n editor capture after Tidy Up. Enlarged branches follow.



FULL CANVAS / ALL 112 NODES

Trigger

A listing inquiry, prospect Telegram message, inbound email or tour event starts the relevant branch.

Key steps and decisions

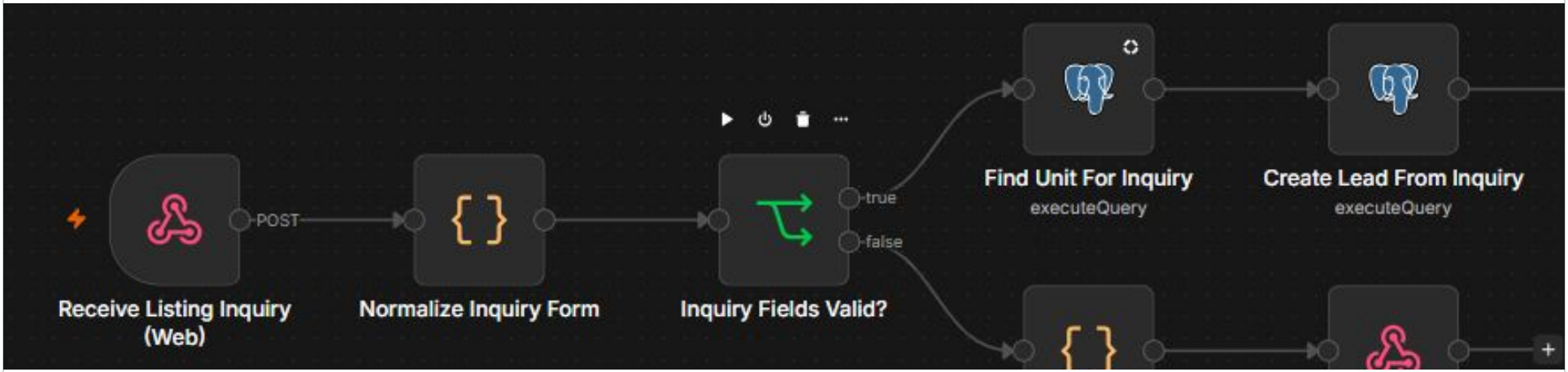
Bind identity by token, dedupe messages, load property facts, extract slots and run fixed prequal and message guardrails.

Output

Logged replies, manager alerts, approved tour requests and application links. The model never decides rental eligibility.

WORKFLOW 01 / INTAKE

The listing becomes a tracked lead



Actual n8n canvas / enlarged for reading

The native listing form posts its unit, name and question into n8n. Field checks reject invalid input; the unit is looked up in Postgres and a lead reference is created. The same unit record supplies both the public listing and the facts the assistant may quote.

WORKFLOW 01 / CONVERSATION

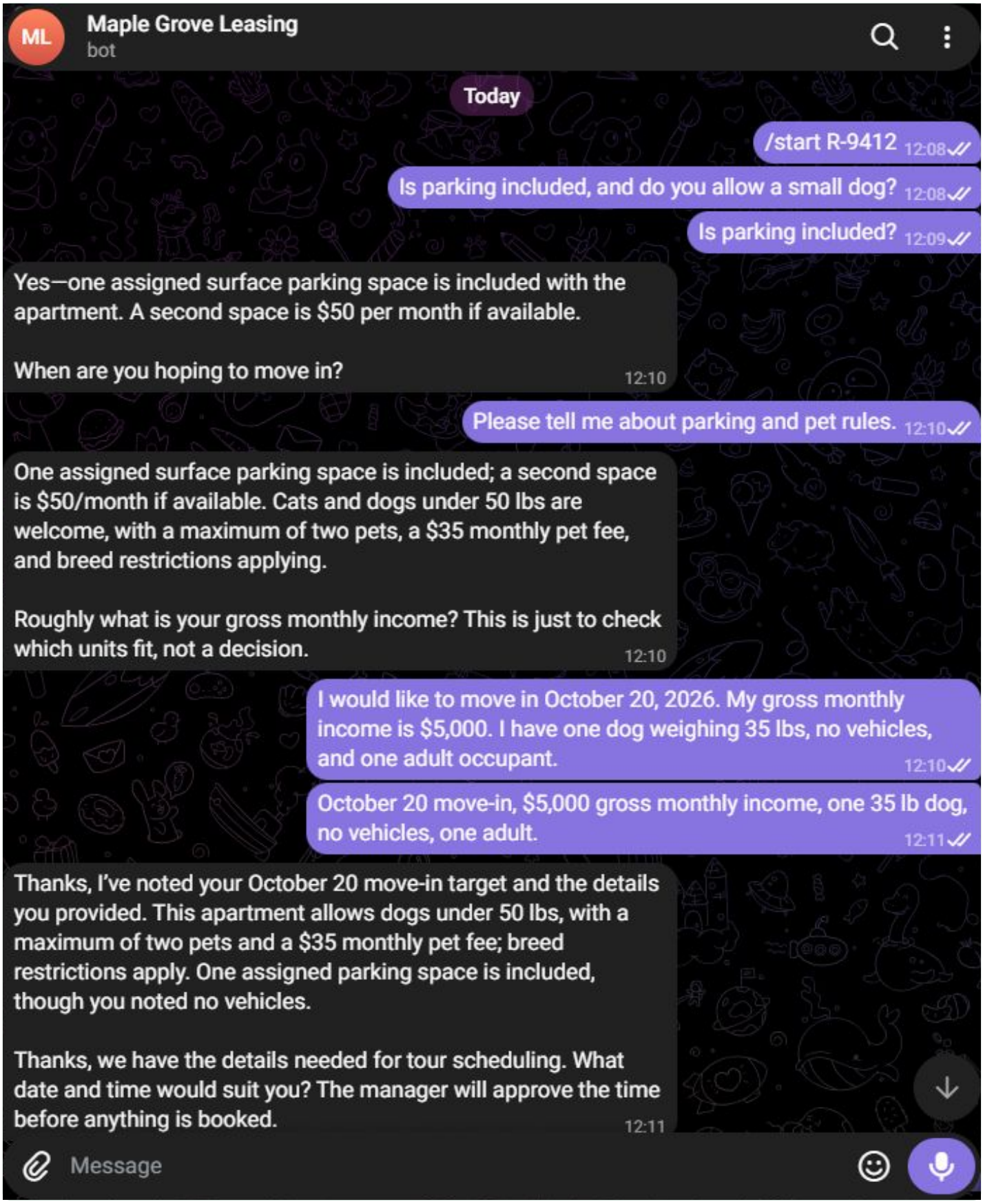
Grounded answers, fixed next steps



Actual n8n canvas / enlarged for reading

The conversation context contains unit facts and recorded answers. OpenAI answers the question, extracts stated fields and classifies intent into structured output. Plain JavaScript merges those slots and selects the next fixed question. Prequalification uses self-reported answers only and is used to coordinate tours.

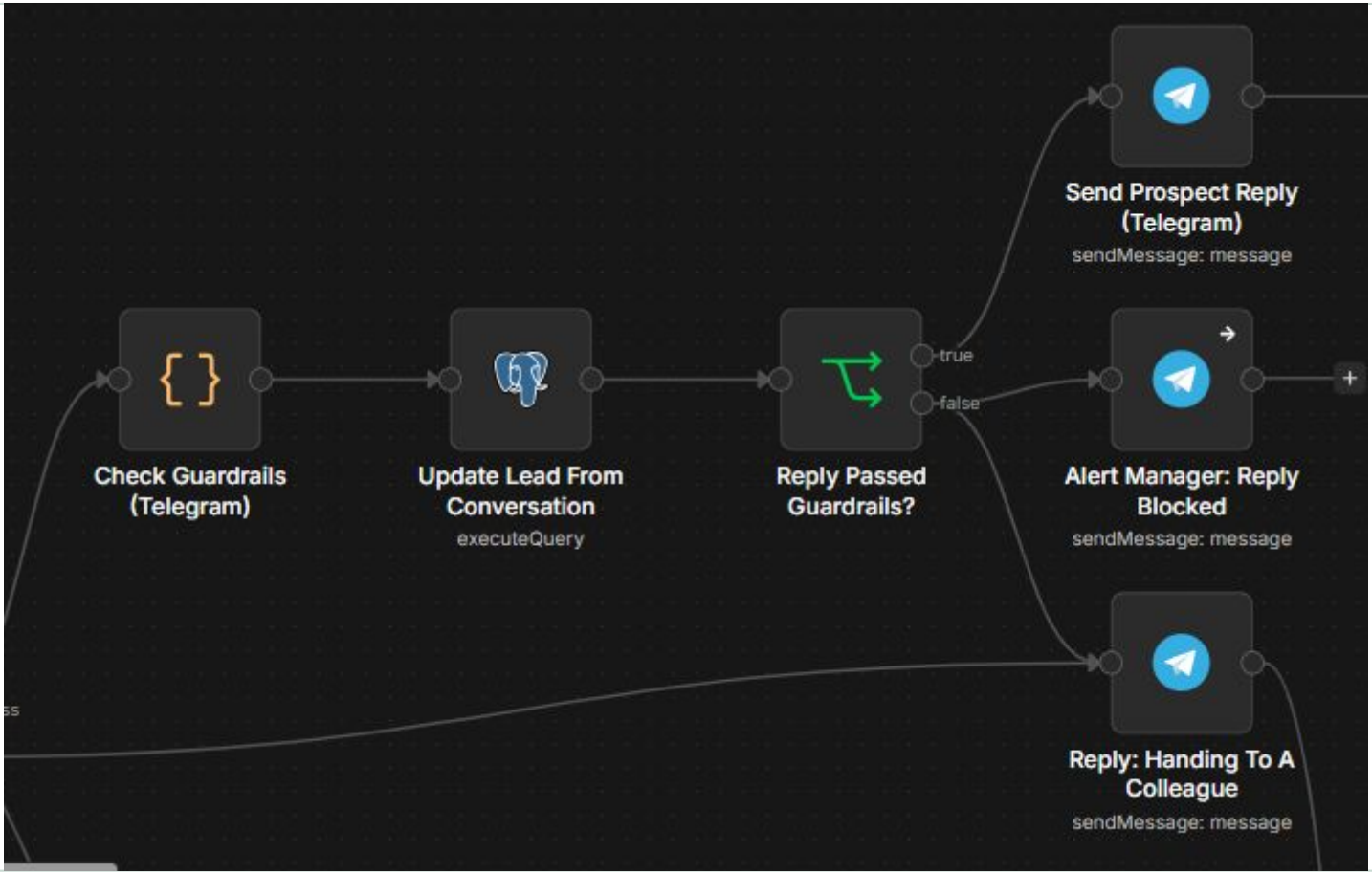
Questions become a tour request



ACTUAL TELEGRAM CHAT / SYNTHETIC RENTER / SIDEBAR EXCLUDED

WORKFLOW 01 / GUARDRAILS

Check the reply before it leaves

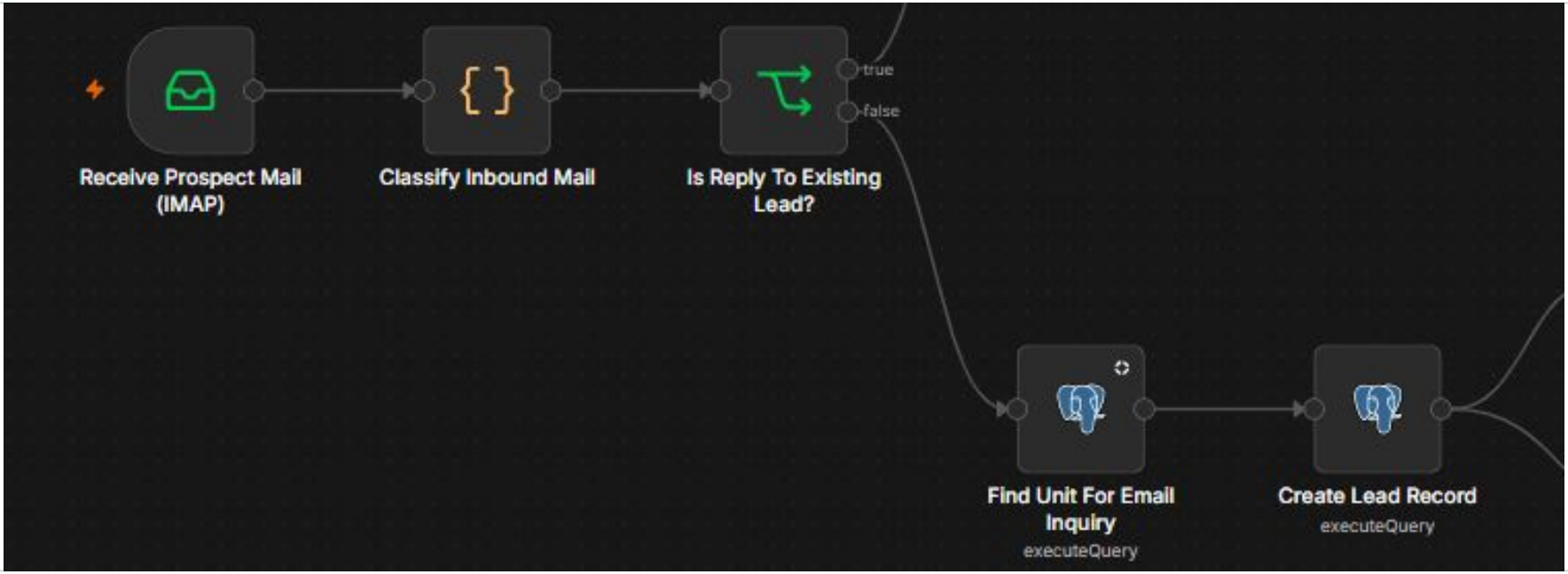


ACTUAL ENLARGED CANVAS / SEND OR HAND OFF

The reply is checked against allowed property facts and the renter's recorded statements. Passing replies are sent and logged. A blocked or failed turn alerts the manager and uses a fixed handoff response, so the renter receives a next step. Screening messages require individual manager approval.

WORKFLOW 01 / EMAIL

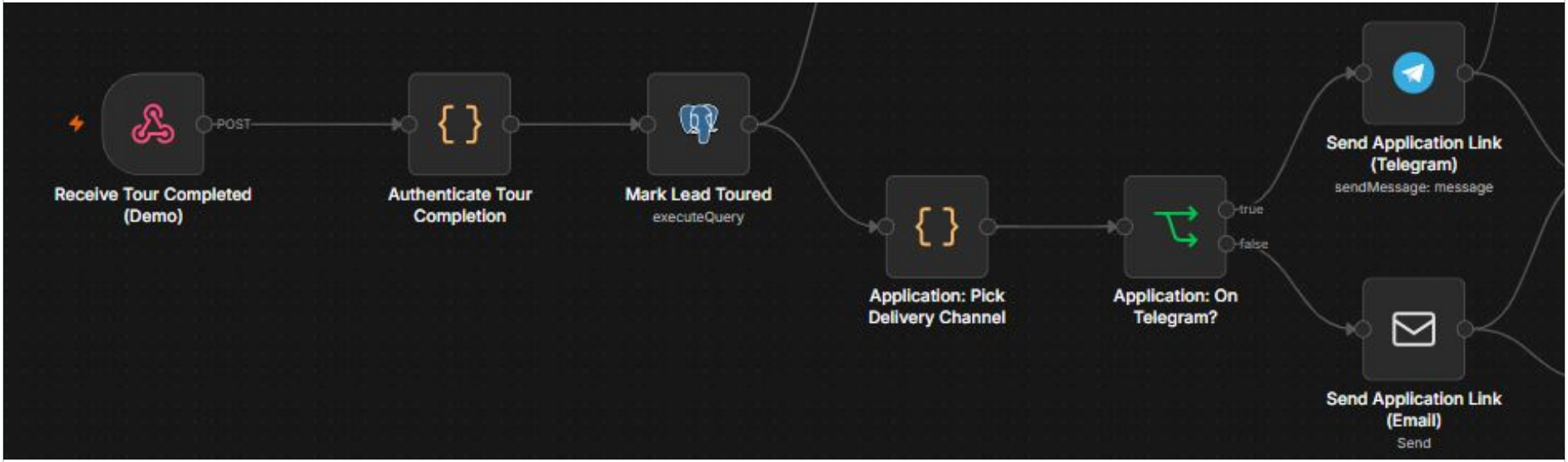
Inbound mail joins the same lead record



Actual n8n canvas / enlarged for reading

Inbound mail is classified and matched to an existing reference token, or becomes a new inquiry. Grounded replies pass factual guardrails; prequalification uses the configured gate. Message IDs prevent duplicate processing. The fresh connected journey in this sample uses Telegram; the email intake branch is shown here as part of the implemented workflow.

A manager authorizes the next step



Actual n8n canvas / enlarged for reading

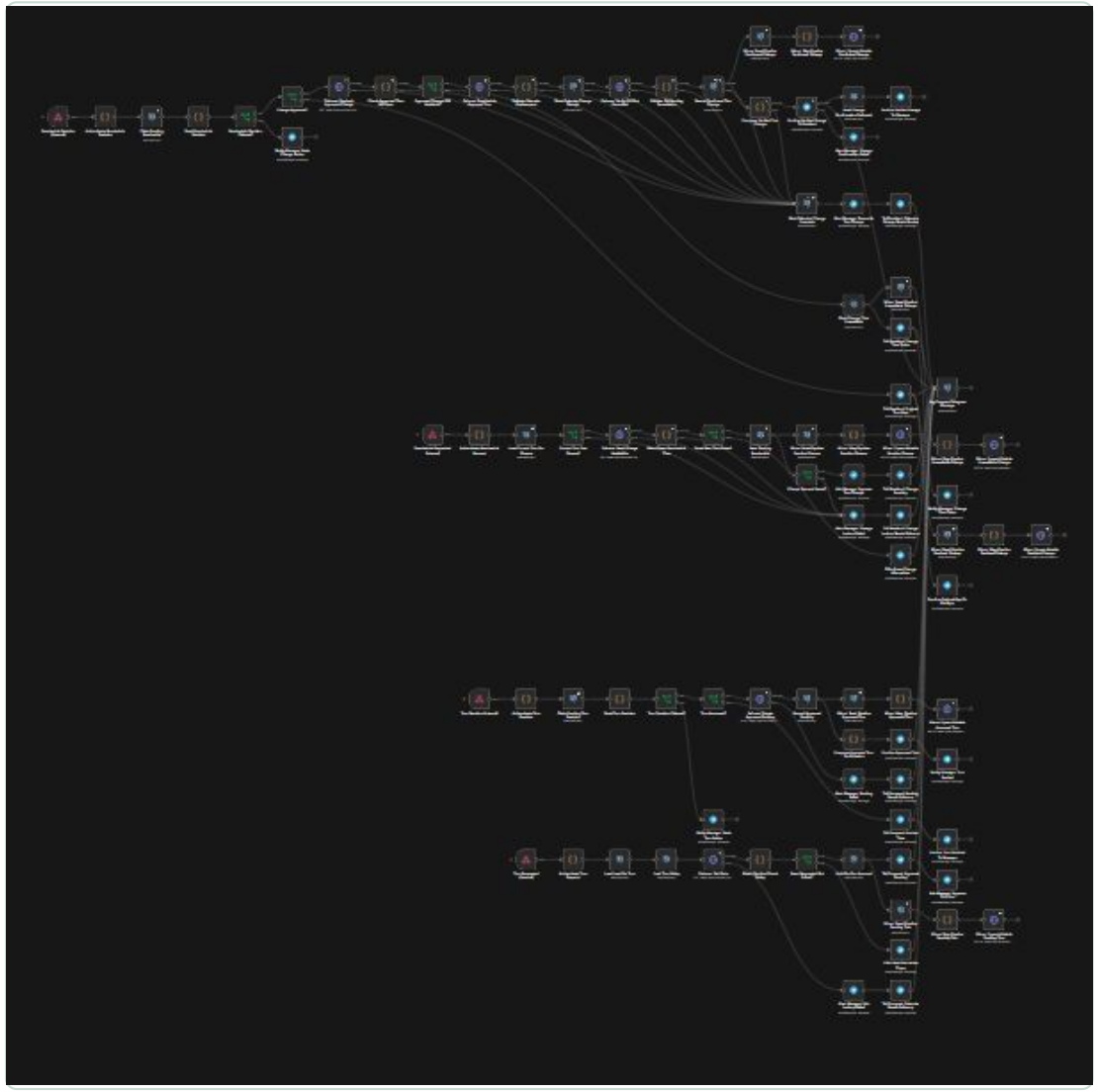
A manager-authorized demo completion marks the lead toured and sends an application link over its recorded channel. The action and message are logged against that lead. The captured attendance is synthetic; it is not evidence of a physical visit.

WORKFLOW 05

Manager-approved tour coordination

Actual n8n editor capture after Tidy Up. Enlarged branches follow.

FULL CANVAS / ALL 91 NODES



Trigger

An identified renter requests an exact time, or asks to change a confirmed tour.

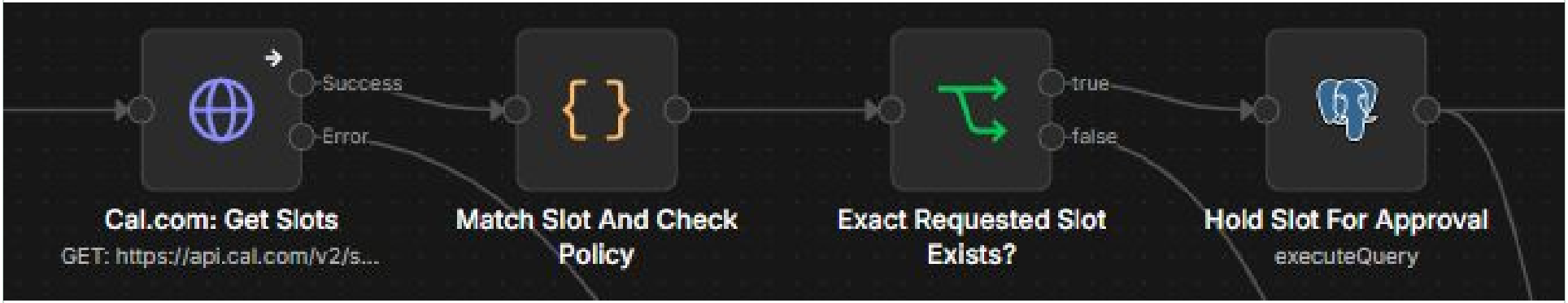
Key steps and decisions

Read Cal.com availability, apply booking policy and hold for a manager. Approval claims the action once before calling the calendar.

Output

A real calendar booking, confirmed time, labelled mock entry code and logged delivery. A change verifies both replacement and original cancellation.

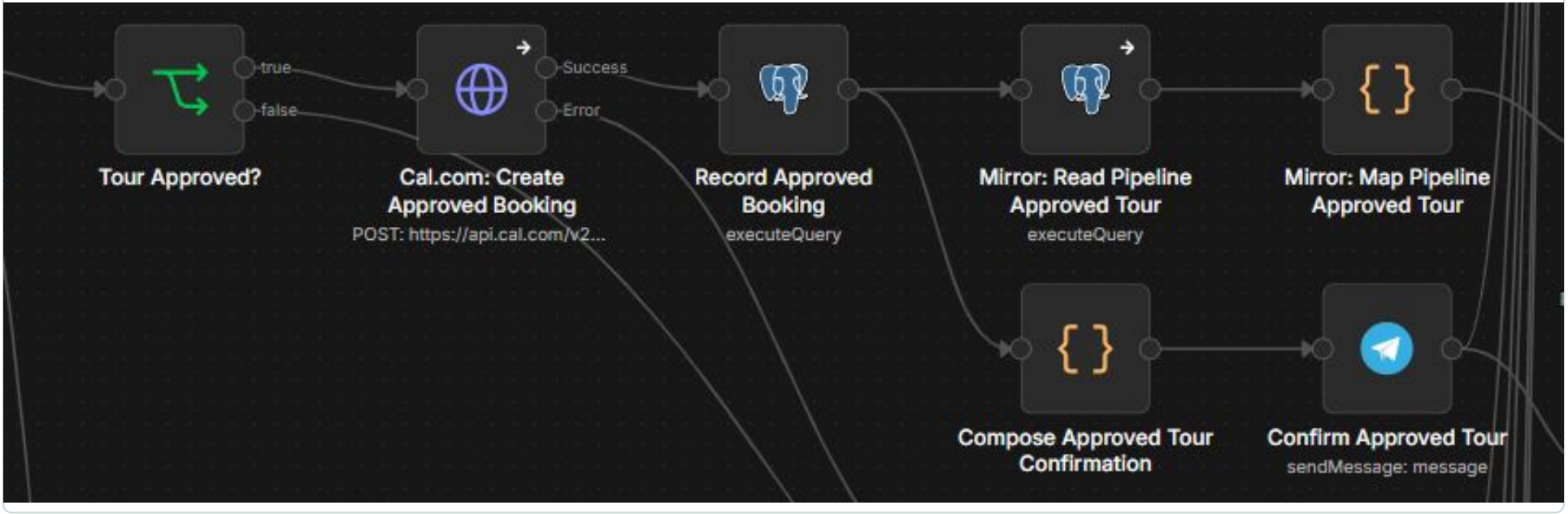
Check the exact requested slot



Actual n8n canvas / enlarged for reading

Cal.com supplies available slots. The requested property-local time must match an actual slot and the configured policy. A valid request produces a pending manager card; approval is required before a booking or entry code is released. The fresh run requested October 12, 2026 at 11:00 AM America/New_York.

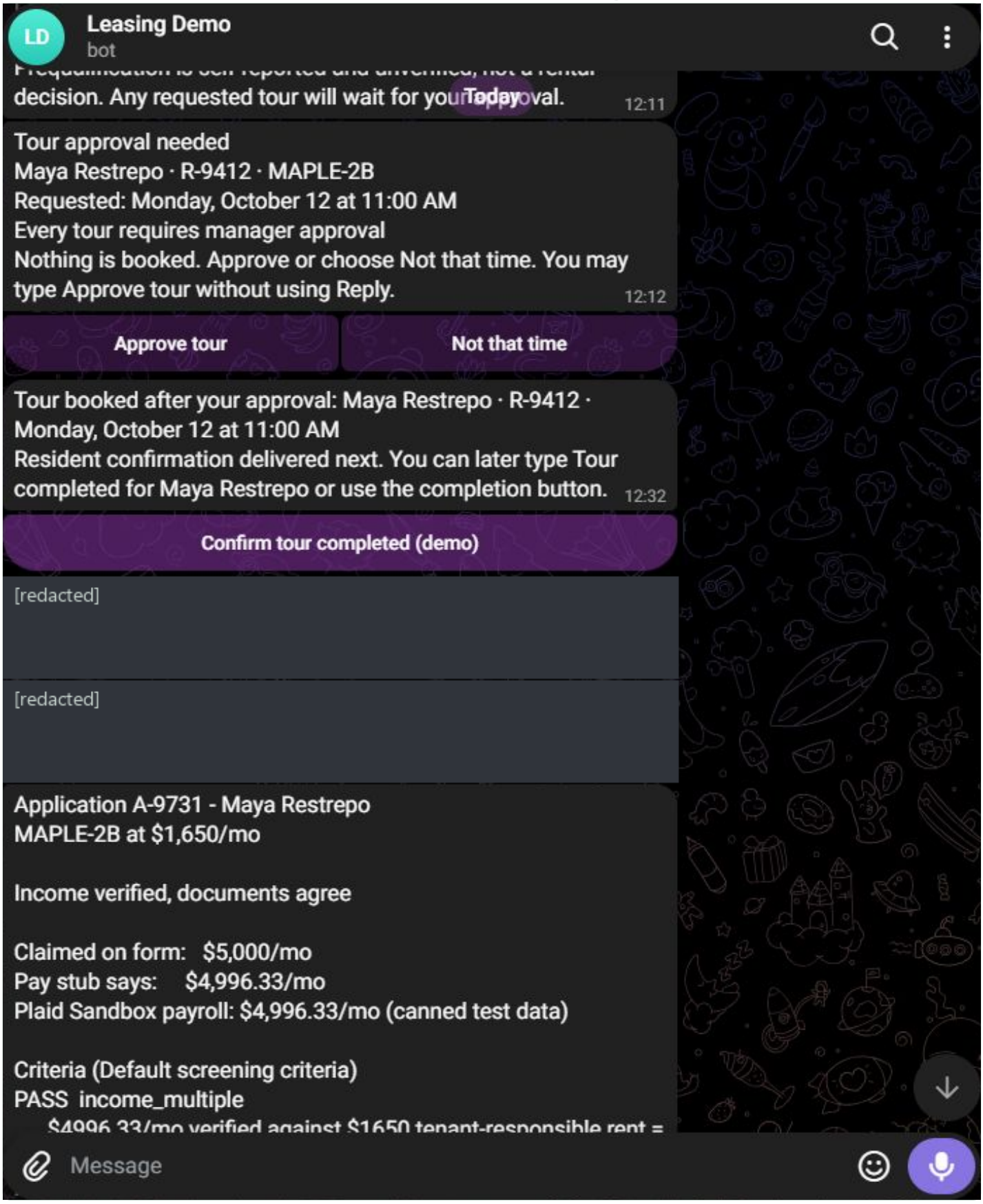
Store and confirm the provider booking



Actual n8n canvas / enlarged for reading

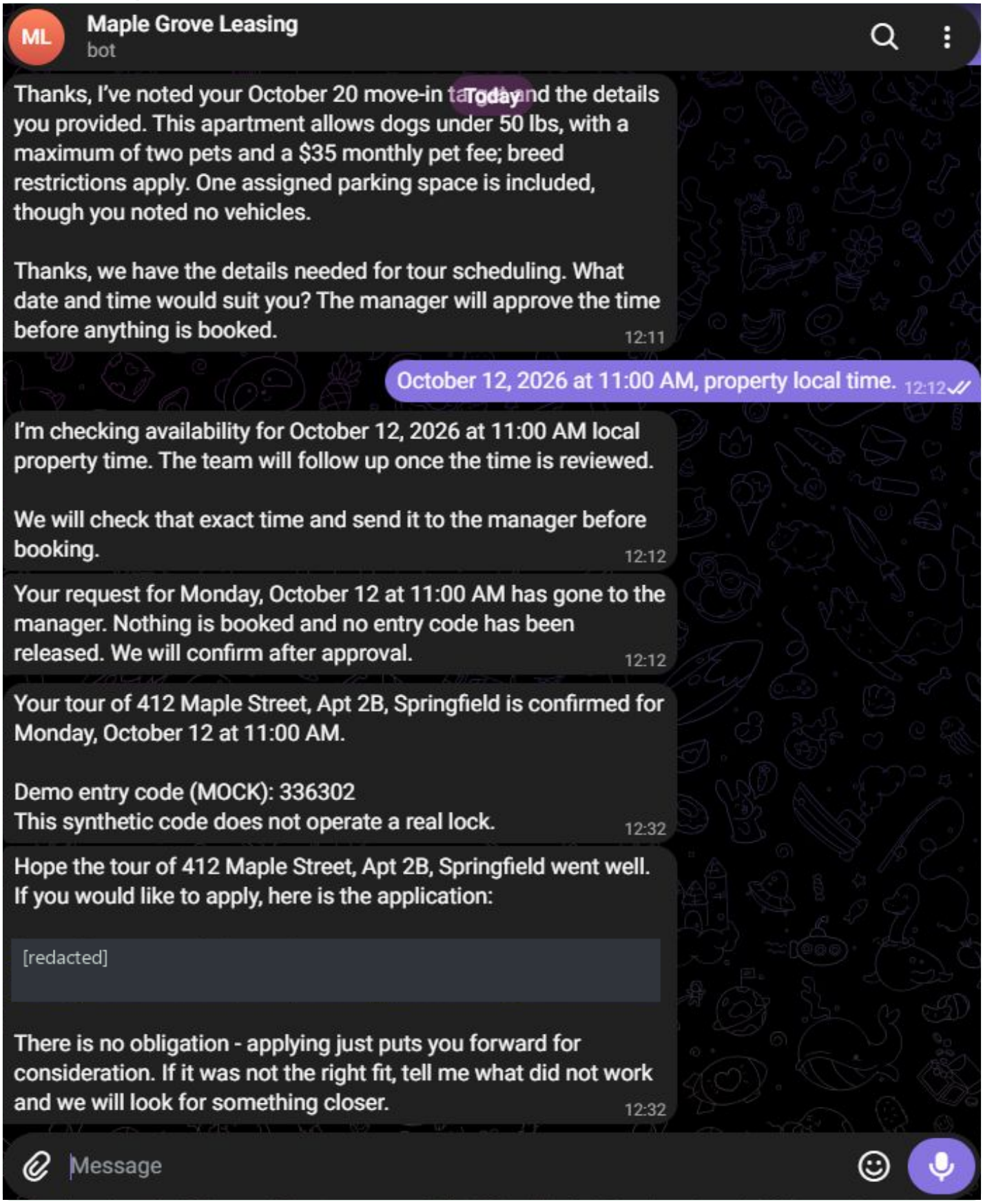
The approval path creates and stores the provider booking, mirrors the phase and composes a confirmation. Provider or delivery failures remain visible to the manager. The following conversation captures show the approval request, booked confirmation and labelled mock entry code for R-9412.

The approval produces a booking



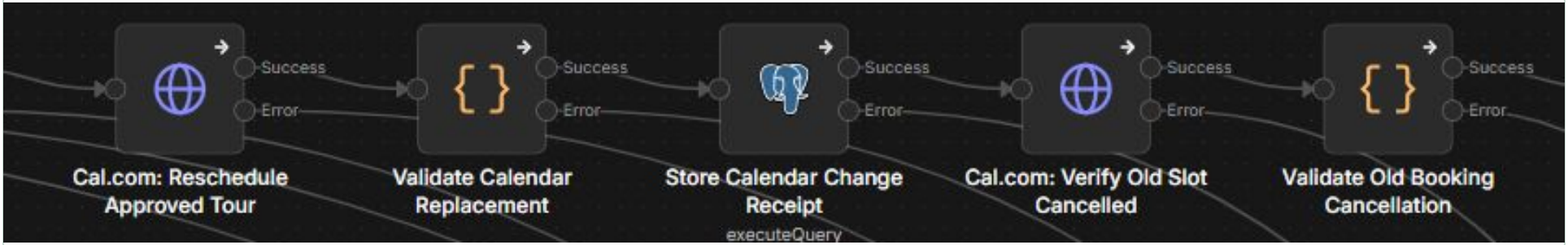
ACTUAL MANAGER CHAT / SYNTHETIC DEMO / PERSONAL DETAILS REDACTED

Pending request to confirmation



ACTUAL RENTER CHAT / MOCK ENTRY CODE / DELIVERY LINK REDACTED

Reschedule with provider verification



Actual n8n canvas / enlarged for reading

A change request preserves the confirmed booking while a manager reviews the replacement time. On approval, the workflow rechecks availability, uses Cal.com's native reschedule operation and stores its receipt. It then verifies the original booking is cancelled before recording and confirming the change. Stale actions cannot claim it twice. This branch is covered by fixture and transactional checks; the fresh screenshots above show the booking path.

WORKFLOW 02

Application intake and evidence checks

Actual n8n editor capture after Tidy Up. Enlarged branches follow.



FULL CANVAS / ALL 24 NODES

Trigger

The renter submits the n8n application form with consent and an uploaded synthetic paystub.

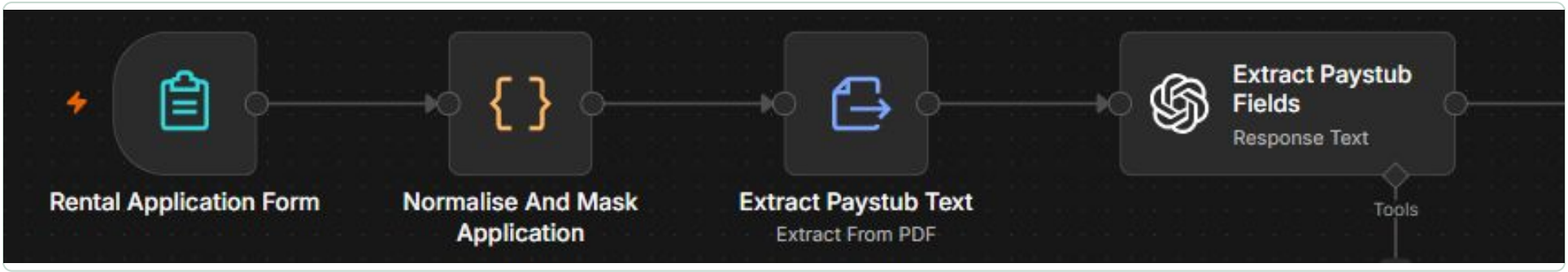
Key steps and decisions

Normalize fields, extract cited document values, call Plaid Sandbox and compare income. Fixed rules evaluate published thresholds.

Output

A manager card shows three income figures and criterion evidence. Files that do not meet criteria stop for human review.

Turn a document into cited facts

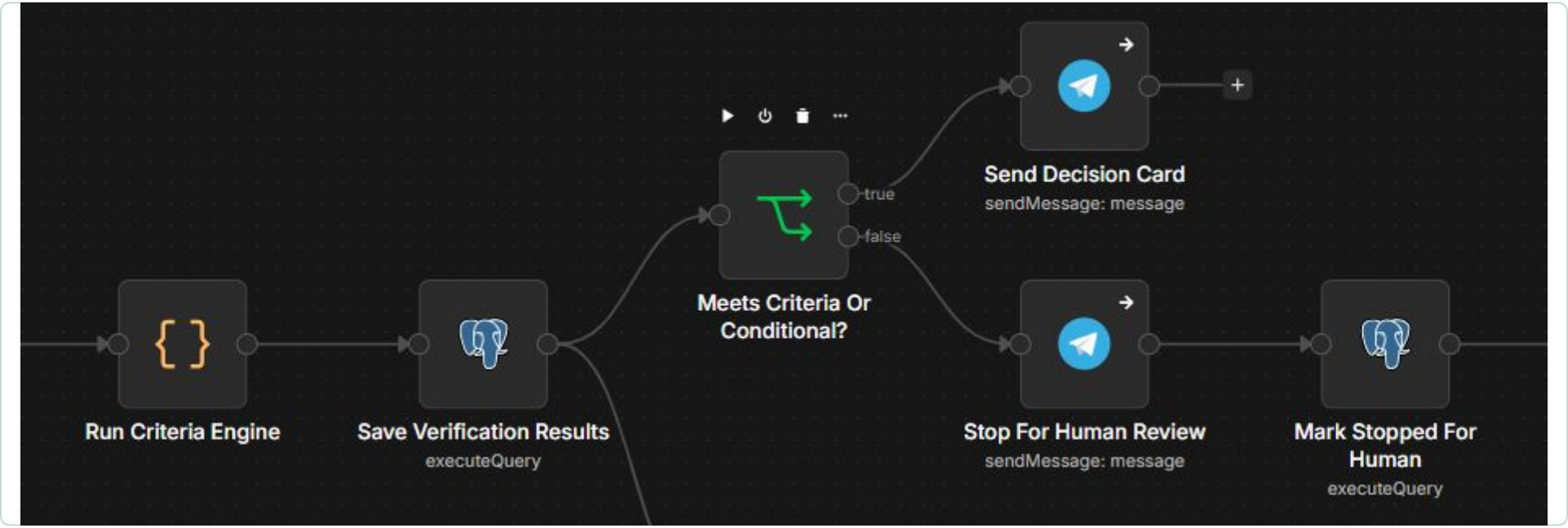


The form normalizes application fields and masks sensitive identifiers on arrival. PDF text extraction precedes database nodes so the uploaded binary survives. OpenAI returns bounded fields and source quotations, including gross pay and pay frequency.



A genuine Plaid Sandbox call returns canned payroll data. Pay-frequency conversion puts claimed, document and Sandbox income on the same monthly basis. Bureau and identity results come from the visibly labelled MOCK adapter.

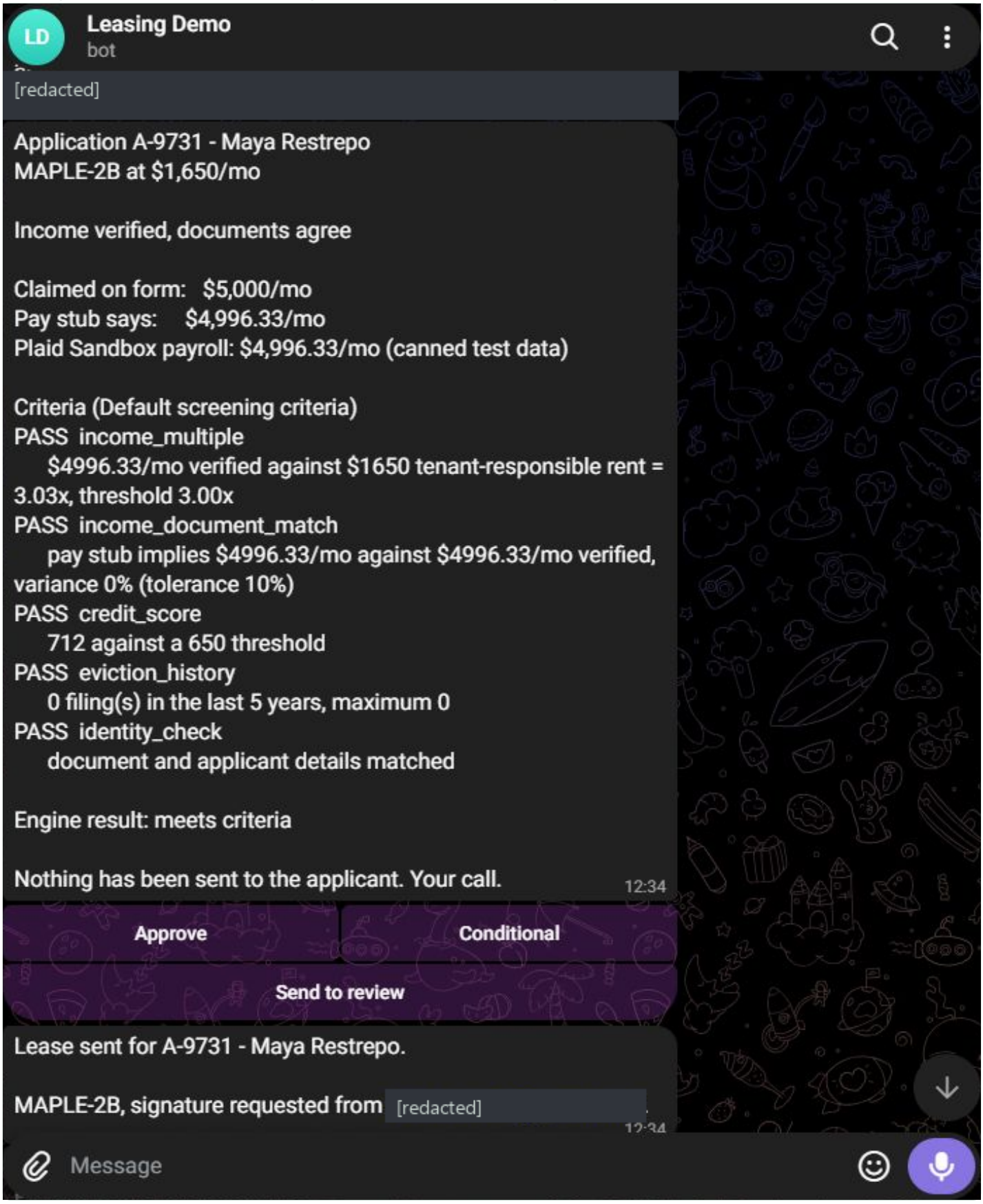
Evidence feeds a deterministic engine



Actual n8n canvas / enlarged for reading

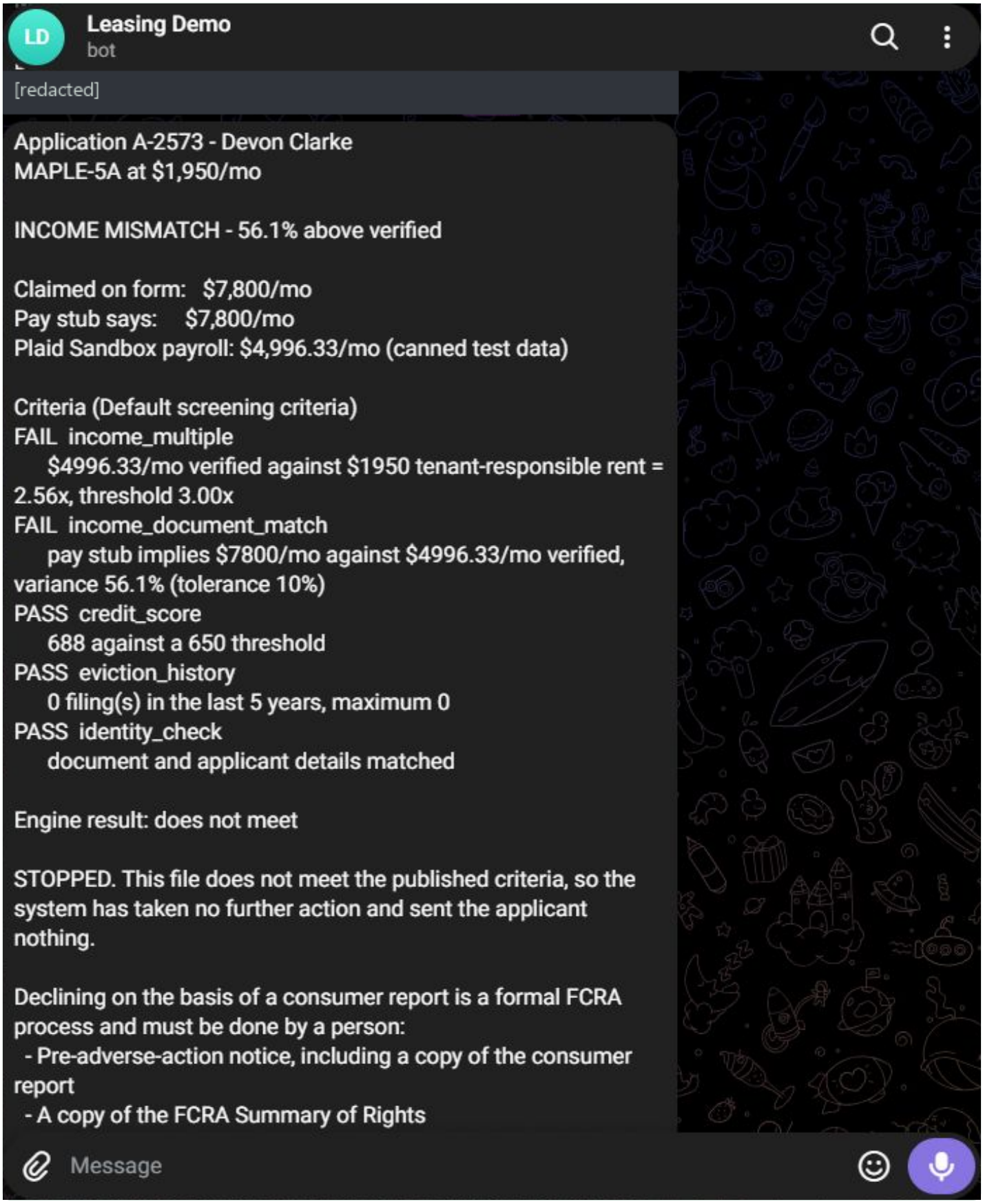
The engine reads manager-configured thresholds and emits a result plus a per-criterion evidence trail. It checks income multiple, document match, synthetic bureau values and mock identity. It calls no model. The manager sees an approval card for meets-criteria or conditional files; does-not-meet files enter the stop-for-human branch.

Maya: facts ready for a manager



ACTUAL MANAGER CHAT / SANDBOX PAYROLL / BUREAU + IDENTITY ARE MOCK

Devon: mismatch stops the flow



ACTUAL MANAGER CHAT / SYNTHETIC APPLICANT / MOCK BUREAU + IDENTITY

Two files, two traceable outcomes

The results below were read from the successful fresh executions and the stored application state.

Monthly figures	Maya / A-9731	Devon / A-2573
Claimed on application	$5,000.00	$7,800.00
Extracted from paystub	$4,996.33	$7,800.00
Plaid Sandbox payroll	$4,996.33	$4,996.33
Income multiple	3.03x against 3.00x	2.56x against 3.00x
Document mismatch	0% / within tolerance	56.1% / above 10%
Engine output	meets_criteria	does_not_meet
Resulting state	Manager approved; lease signed	Stopped for human review

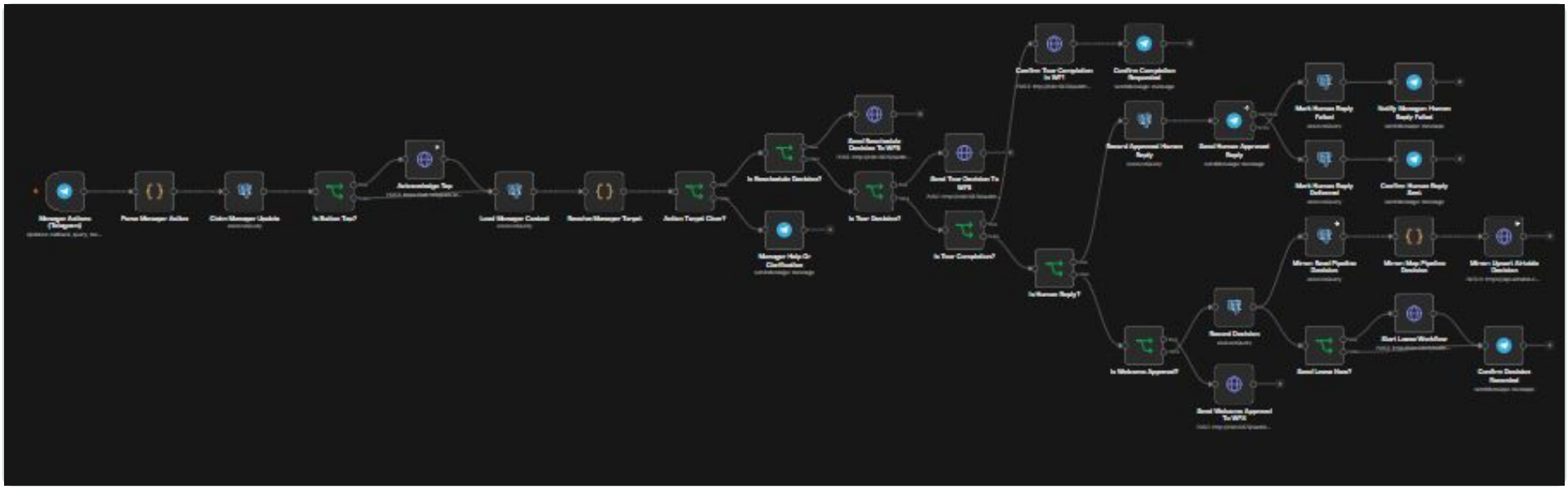
The stop is observable

Devon's successful execution took the stop branch. No lease was created and no applicant-facing outcome message was sent. The manager retains control of the next action.

WORKFLOW 03

Manager actions and approvals

Actual n8n editor capture after Tidy Up. Enlarged branches follow.



FULL CANVAS / ALL 32 NODES

Trigger

An inline button or manager command arrives through the dedicated manager Telegram bot.

Key steps and decisions

Parse and dedupe the update, resolve the target and route tours, changes, completion, replies, application decisions or welcome approval.

Output

An actor and timestamp are recorded. Only the appropriate approved action is dispatched to the owning workflow.

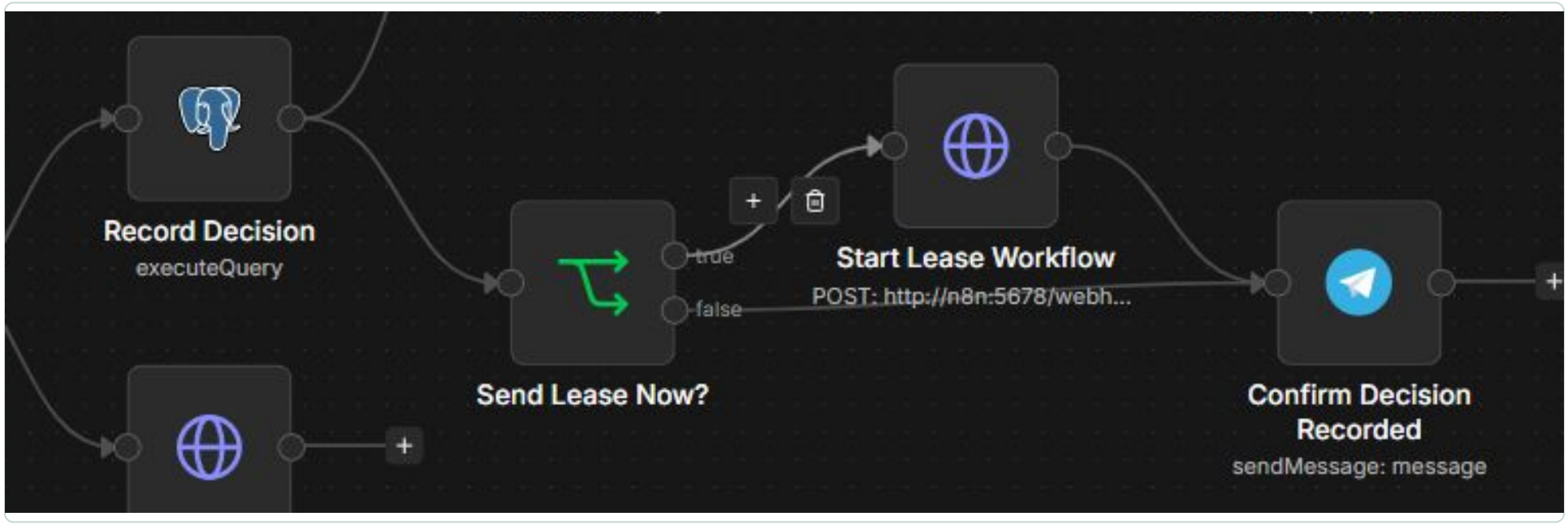
One action, one identified record



Actual n8n canvas / enlarged for reading

The router claims the incoming update and distinguishes button taps from text commands. It loads the current manager context, resolves the record and asks for clarification if the target is ambiguous. Tour approval, tour change, synthetic completion, approved replies, application decisions and welcome approval each have their own route.

Record approval before starting the lease



Actual n8n canvas / enlarged for reading

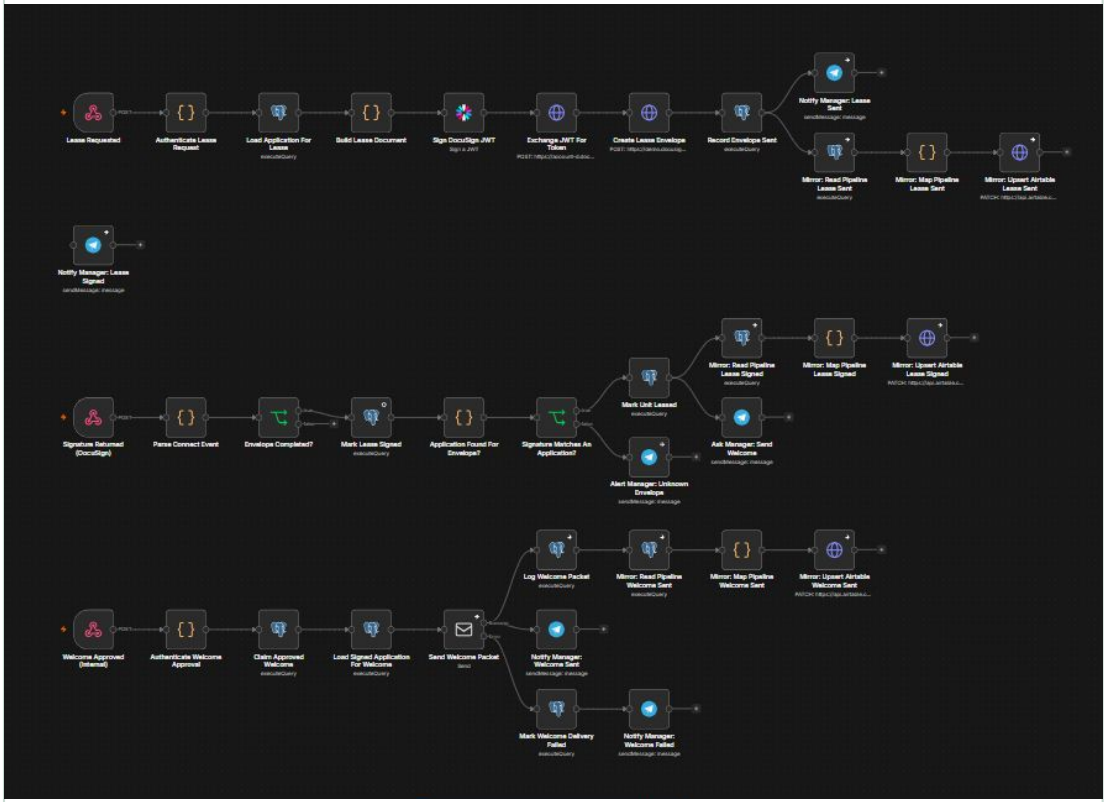
Application decisions are persisted before the lease gate is evaluated. Approved files can enter WF4; review and conditional actions remain controlled by the manager. The fresh run recorded an approved decision for A-9731, then started its sandbox lease. The engine's meets-criteria label alone cannot create a lease.

WORKFLOW 04

Lease, signature callback and welcome

Actual n8n editor capture after Tidy Up. Enlarged branches follow.

FULL CANVAS / ALL 37 NODES



Trigger

An approved application starts document creation. DocuSign completion and an approved welcome each enter their own branch.

Key steps and decisions

Fill a fixed template, create the sandbox envelope and match its callback deterministically. Mark the unit leased only for a matched completion.

Output

Stored envelope and signature status, a manager welcome-approval card, then a logged SMTP send after that separate approval.

A fixed document enters DocuSign

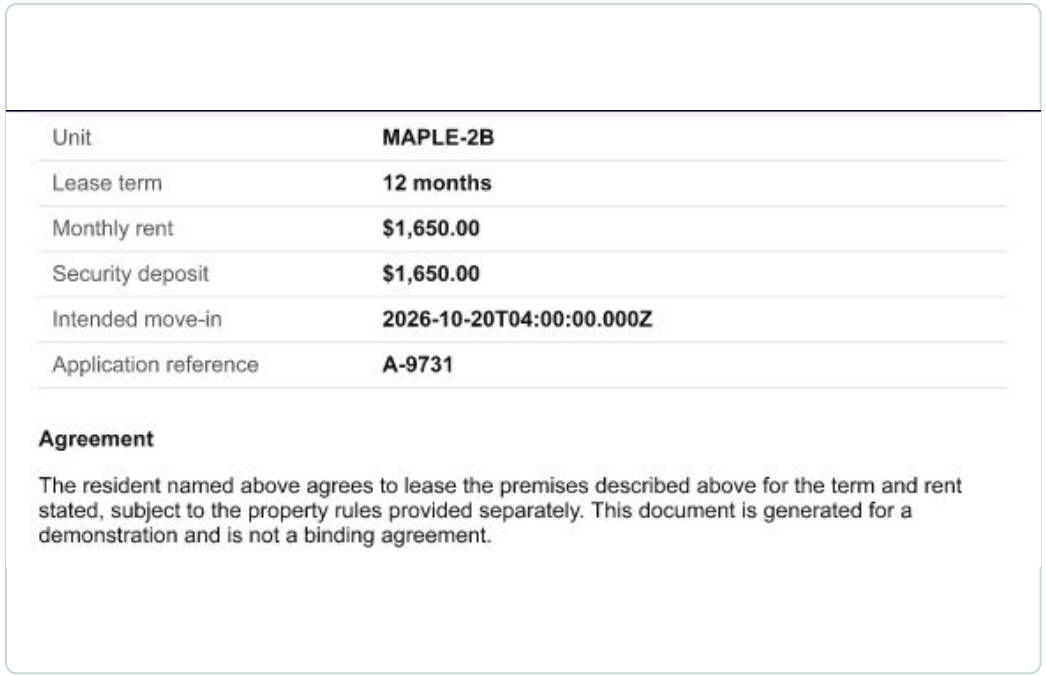


Actual n8n canvas / enlarged for reading

WF4 loads the approved application and unit facts, fills a fixed HTML template and sends it to the DocuSign developer sandbox for PDF conversion and signing. The model writes no lease clauses or rent figures. The envelope ID and sent state are recorded; the invitation was received in the approved demo mailbox.

Nonbinding sandbox document completed

The document and completion screen below are actual captures from the fresh signing session.



Unit	**MAPLE-2B**
Lease term	**12 months**
Monthly rent	**$1,650.00**
Security deposit	**$1,650.00**
Intended move-in	**2026-10-20T04:00:00.000Z**
Application reference	**A-9731**

Agreement

The resident named above agrees to lease the premises described above for the term and rent stated, subject to the property rules provided separately. This document is generated for a demonstration and is not a binding agreement.



Maya Restrepo is a fictional signer. The document explicitly states that it is generated for demonstration and is not a binding agreement.

The provider's completion screen is paired with the actual callback execution on the next page. This links the signing action to the application's recorded state.

The provider event updates the record



The callback responds immediately. It parses the Connect event, checks for envelope completion and resolves the envelope to its stored application. A matched signature updates the lease and unit state; an unknown envelope alerts the manager.

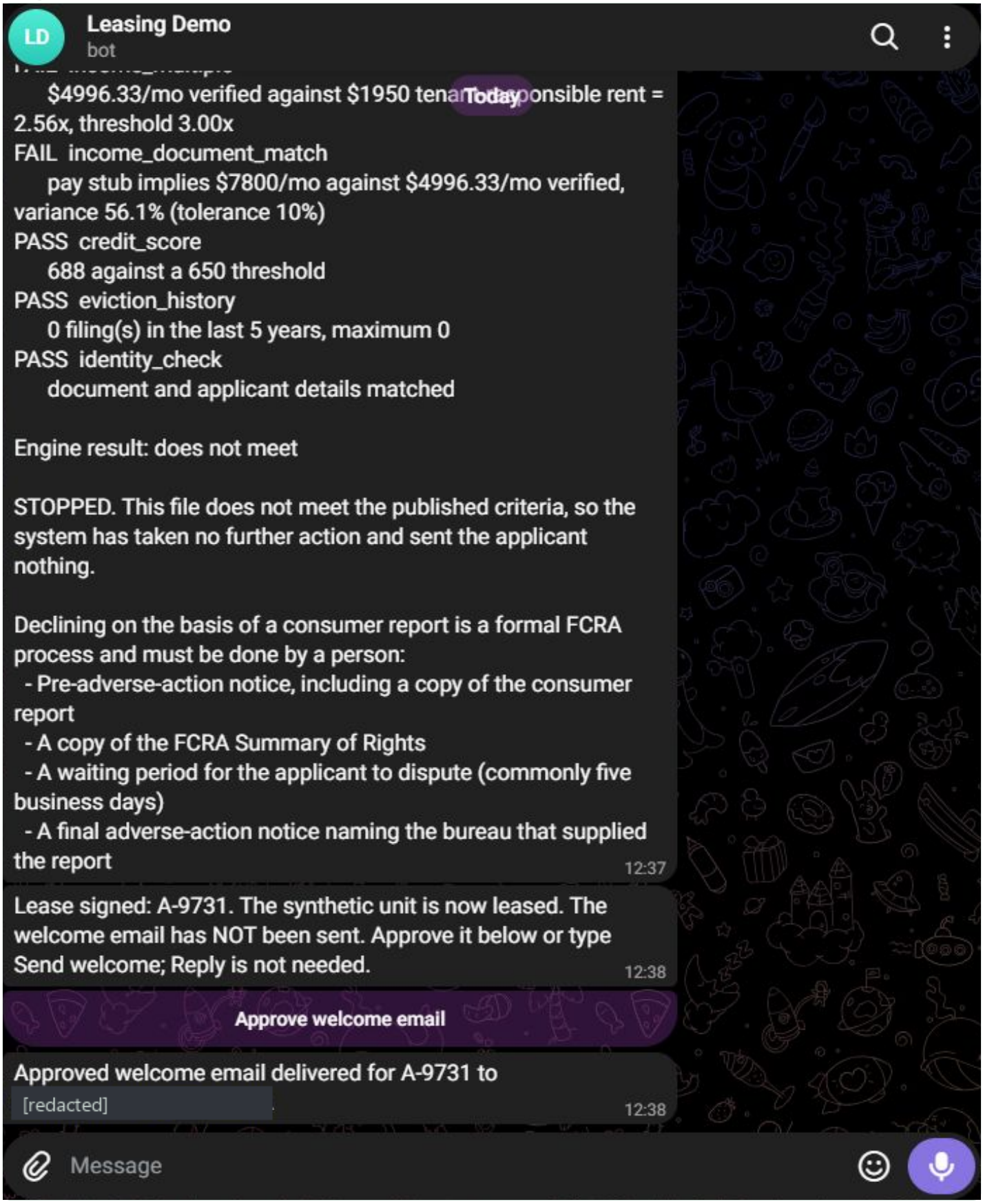


ACTUAL N8N EXECUTION / SIGNATURE CALLBACK #321

Signing does not send the welcome automatically

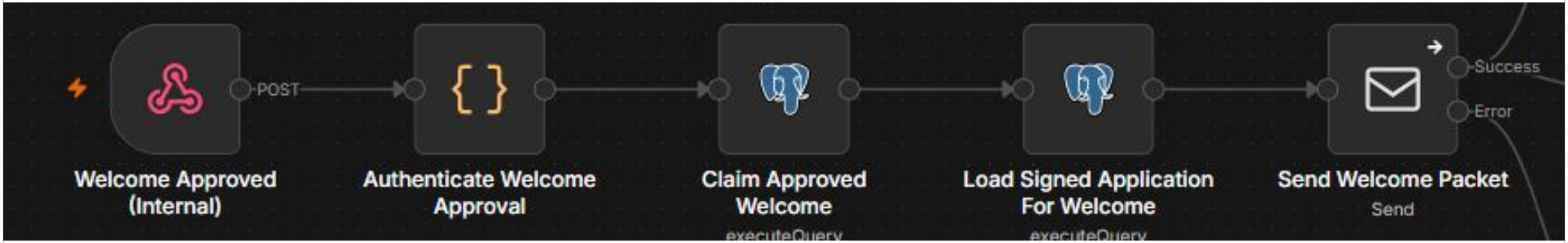
The recorded completion creates a separate manager card. The welcome email waits for that approval.

Signature, stop case and welcome



ACTUAL MANAGER CHAT / SYNTHETIC DATA / PERSONAL DETAILS REDACTED

An approved welcome was received



The internal approval is authenticated and claimed once. WF4 loads the signed application, sends the fixed welcome packet over SMTP and records delivery. Failure takes a visible manager-alert path rather than a successful-send log.



ACTUAL N8N EXECUTION / WELCOME DELIVERY #323

Receipt verified separately

The demo mailbox contained the welcome message carrying application reference A-9731. The received invitation, provider completion, callback and welcome receipt were checked separately.

VERIFICATION

Evidence behind the screenshots

Each row below was checked against the actual fresh n8n execution, returned output or received message.

307 / Grounded reply	Parking and pet facts answered in the prospect chat.
311 + 313 / Tour	Exact slot held for approval; approved Cal.com booking recorded and confirmed.
315 + 316 / Follow-up	Manager-recorded synthetic attendance authorized the application link.
317 / Application	Uploaded paystub, cited extraction, Sandbox payroll and criterion evidence completed.
318 + 319 / Approval + lease	Manager decision recorded; actual sandbox envelope created.
320 / Stop case	Devon mismatch produced does_not_meet and stopped_for_human.
321 / Signature	Genuine provider callback matched A-9731; lease signed and unit leased.
322 + 323 / Welcome	Separate approval, successful SMTP send and received welcome message.

Additional behavior checks

146 local checks passed across screening, conversation, manager routing, application form, pipeline mirror and tour rescheduling. These checks support branch behavior; the screenshots document the fresh live demo.